Merisa Anggi Anggraini

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Summary

Mathematics graduate from Institut Teknologi Sepuluh Nopember (ITS) with strong competencies in data analytics, machine learning, and software development. Experienced in executing applied AI projects, including deep learning implementations for computer vision and statistical time-series forecasting for real-world data analysis. Proficient in Python, Java, database management, and UI design, complemented by proven leadership, event management, and partnership experience across multiple student organizations and tech communities.

Experience

Digistar Club ITS by Telkom Indonesia 2024 – 2026
Partnership Staff *Surabaya, Indonesia*

- Initiated and managed strategic outreach to external partners, educational institutions, and student tech communities.
- Coordinated talent development initiatives and digital workshops to enhance community engagement and technical competencies.
- Collaborated with cross-functional teams to design promotional strategies and streamline partnership agreements.

PT TASPEN (Persero) - Kantor Pusat 2025
Research & Development Analyst Intern – Strategic Planning Division *Jakarta, Indonesia*

- Conducted time-series analysis and predictive modeling on national Death Benefit (JKM) claim datasets using the SARIMA method.
- Performed data preprocessing, stationarity diagnostics (ADF test), and seasonal differencing to address trend and weekly seasonality patterns.
- Optimized model hyperparameters through grid search evaluation (AIC/BIC) and validated residuals to ensure forecast reliability and model stability.
- Developed and implemented a web-based prototype (*JKM Forecast System*) to visualize 30-day future claim projections and support strategic decision-making.

OMITS 17th (Olimpiade Matematika ITS) 2024
Sub-division Head of Administration & Communication (Atcom) *Surabaya, Indonesia*

- Led the administrative workflow and external correspondence for a nationwide mathematics competition involving thousands of participants.
- Coordinated communication channels between committee divisions, regional supervisors, and participating institutions.

HIMATIKA ITS Jan 2024 – Dec 2024
Student Welfare Staff *Surabaya, Indonesia*

- Created and managed Student Welfare (SW) news and information to support student communication and engagement.

UKM Robotika ITS 2023 – 2024
Human Resource Development (PSDM) Staff *Surabaya, Indonesia*

- Facilitated member recruitment, talent screening, and onboarding programs for robotics research divisions.
- Organized internal soft-skill and technical training sessions to improve team workflow and project execution.
- Monitored member performance and organizational health across collegiate competition workstreams.

ILITS Forda Jember Nov 2023 – Feb 2024
Event Division Staff *Jember, Indonesia*

- Created the event rundown for INI LHO ITS 2024 to ensure a structured and well-coordinated event flow.
- Supervised event proceedings and coordinated activities to ensure smooth execution according to the planned rundown.

IFC (ITS FUTSAL CHAMPIONSHIP) Aug 2023 – Oct 2023
Event Division Staff *Surabaya, Indonesia*

- Assisted in event planning and coordinated activities to ensure smooth tournament operations.
- Coordinated Match Day activities between teams, referees, and relevant committees.

- Managed ceremonial segments and ensured activities followed the planned event schedule.
- Served as a liaison between teams, referees, and committees to ensure smooth flow and punctuality of activities.

OMITS 16th (Olimpiade Matematika ITS) Jun 2023 – May 2024
Staff, Partner Accountants Division *Surabaya, Indonesia*
- Identified and coordinated accommodation and transportation options to support event operations.
- Assisted in coordinating logistical requirements with relevant parties and external service providers.

The Beauty Show Volunteer Jul 2023
Equipment Division Staff *Surabaya, Indonesia*
- Prepared and provided equipment and logistical needs to support the successful execution of the event.

Education

Institut Teknologi Sepuluh Nopember (ITS) Aug 2022 - Aug 2026
Bachelor of Mathematics
- Skills: Linear algebra, Algorithm, Mathematical Logic, Data Science

Skills

Programming Languages: Python, Java, SQL
Data Science & AI: Deep Learning, Computer Vision (Image Restoration), Time-Series Forecasting, Data Mining, Predictive Modeling
DevOps & Backend: FastAPI, REST API Development, Podman, Docker, Kubernetes/OpenShift, Ansible, MongoDB
Tools & Technologies: PyTorch, MySQL/Relational Databases, UI/UX Design (Figma), Decision Support Systems, Minitab, SPSS, Excel
Applied Mathematics: Linear Algebra, Numerical Methods, Probability & Statistics, Cryptography, Data Structures
Soft Skills: Cross-functional Collaboration, Partnership Management, Team Leadership, Public Communication, Problem Solving

Language

Indonesian (Native Language) • Javanese (Native Language) • English (Intermediate level)

Projects

Financial Tracker Microservice – Bootcamp ITS Final Project Sep 2026
- Built and deployed a two-service financial tracking application (FastAPI) featuring transaction management and automated monthly spending-alert profiling.
- Containerized services with Podman and deployed them to Red Hat OpenShift, configuring inter-service networking, environment-based secrets, and multi-replica deployments.
- Provisioned a self-hosted MongoDB instance on a virtual machine and automated its setup using Ansible through Ansible Automation Platform (AAP), with source control managed on GitHub.

Single Image Deraining via Frequency-Guided Deep Unfolding (RCDNet) Jul 2026
- Engineered a deep unfolding neural network integrating a Spectral Gate Bottleneck (*FrequencyHint*) to capture global frequency features for single-image rain streak removal.
- Optimized model training workflows across synthetic and real-world datasets, achieving measurable gains in PSNR/SSIM and a 13.82% improvement in perceptual quality (BRISQUE score) on real outdoor images

JKM Claims Predictive Analytics & Forecasting Web System Sep 2025
- Developed an end-to-end time-series forecasting pipeline using SARIMA to model daily Death Benefit (JKM) claim volumes, handling trend and weekly seasonality.
- Automated model parameter optimization via grid search (AIC/BIC selection) and deployed a prototype web dashboard to deliver 30-day interactive claim projections for operational decision support

Cryptographic Security Implementation: RSA & ElGamal 2025
- Implemented RSA and ElGamal public-key cryptographic algorithms to demonstrate key generation, encryption, and decryption processes.
- Developed and tested cryptographic workflows to analyze the implementation and security characteristics of asymmetric encryption techniques.

Distributed Database System 2025
- Designed and implemented a distributed database system to manage and process data across multiple database nodes.
- Applied distributed database concepts including data distribution, replication, and query processing to improve data availability and system scalability.